
Physics C: Electricity and Magnetism

Practice Exam

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ADVANCED PLACEMENT PHYSICS C TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS	
Proton mass, $m_p = 1.67 \times 10^{-27}$ kg	Electron charge magnitude, $e = 1.60 \times 10^{-19}$ C
Neutron mass, $m_n = 1.67 \times 10^{-27}$ kg	1 electron volt, $1 \text{ eV} = 1.60 \times 10^{-19}$ J
Electron mass, $m_e = 9.11 \times 10^{-31}$ kg	Speed of light, $c = 3.00 \times 10^8$ m/s
Avogadro's number, $N_0 = 6.02 \times 10^{23} \text{ mol}^{-1}$	Universal gravitational constant, $G = 6.67 \times 10^{-11} (\text{N}\cdot\text{m}^2)/\text{kg}^2$
Universal gas constant, $R = 8.31 \text{ J}/(\text{mol}\cdot\text{K})$	Acceleration due to gravity at Earth's surface, $g = 9.8 \text{ m/s}^2$
Boltzmann's constant, $k_B = 1.38 \times 10^{-23} \text{ J/K}$	
1 unified atomic mass unit,	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg} = 931 \text{ MeV}/c^2$
Planck's constant,	$h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s} = 4.14 \times 10^{-15} \text{ eV}\cdot\text{s}$
	$hc = 1.99 \times 10^{-25} \text{ J}\cdot\text{m} = 1.24 \times 10^3 \text{ eV}\cdot\text{nm}$
Vacuum permittivity,	$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$
Coulomb's law constant, $k = 1/(4\pi\epsilon_0) = 9.0 \times 10^9 (\text{N}\cdot\text{m}^2)/\text{C}^2$	
Vacuum permeability,	$\mu_0 = 4\pi \times 10^{-7} (\text{T}\cdot\text{m})/\text{A}$
Magnetic constant, $k' = \mu_0/(4\pi) = 1 \times 10^{-7} (\text{T}\cdot\text{m})/\text{A}$	
1 atmosphere pressure,	$1 \text{ atm} = 1.0 \times 10^5 \text{ N/m}^2 = 1.0 \times 10^5 \text{ Pa}$

UNIT SYMBOLS	meter, m	mole, mol	watt, W	farad, F
	kilogram, kg	hertz, Hz	coulomb, C	tesla, T
	second, s	newton, N	volt, V	degree Celsius, °C
	ampere, A	pascal, Pa	ohm, Ω	electron volt, eV
	kelvin, K	joule, J	henry, H	

PREFIXES		
Factor	Prefix	Symbol
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin \theta$	0	$1/2$	$3/5$	$\sqrt{2}/2$	$4/5$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$4/5$	$\sqrt{2}/2$	$3/5$	$1/2$	0
$\tan \theta$	0	$\sqrt{3}/3$	$3/4$	1	$4/3$	$\sqrt{3}$	∞

The following assumptions are used in this exam.

- I. The frame of reference of any problem is inertial unless otherwise stated.
- II. The direction of current is the direction in which positive charges would drift.
- III. The electric potential is zero at an infinite distance from an isolated point charge.
- IV. All batteries and meters are ideal unless otherwise stated.
- V. Edge effects for the electric field of a parallel plate capacitor are negligible unless otherwise stated.

ADVANCED PLACEMENT PHYSICS C EQUATIONS

MECHANICS

$$v_x = v_{x0} + a_x t$$

$$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$$

$$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$\vec{J} = \int \vec{F} dt = \Delta \vec{p}$$

$$\vec{p} = m\vec{v}$$

$$|\vec{F}_f| \leq \mu |\vec{F}_N|$$

$$\Delta E = W = \int \vec{F} \cdot d\vec{r}$$

$$K = \frac{1}{2} m v^2$$

$$P = \frac{dE}{dt}$$

$$P = \vec{F} \cdot \vec{v}$$

$$\Delta U_g = mg \Delta h$$

$$a_c = \frac{v^2}{r} = \omega^2 r$$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\alpha} = \frac{\sum \vec{\tau}}{I} = \frac{\vec{\tau}_{net}}{I}$$

$$I = \int r^2 dm = \sum m r^2$$

$$x_{cm} = \frac{\sum m_i x_i}{\sum m_i}$$

$$v = r \omega$$

$$\vec{L} = \vec{r} \times \vec{p} = I \vec{\omega}$$

$$K = \frac{1}{2} I \omega^2$$

$$\omega = \omega_0 + \alpha t$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

$$a = \text{acceleration}$$

$$E = \text{energy}$$

$$F = \text{force}$$

$$f = \text{frequency}$$

$$h = \text{height}$$

$$I = \text{rotational inertia}$$

$$J = \text{impulse}$$

$$K = \text{kinetic energy}$$

$$k = \text{spring constant}$$

$$\ell = \text{length}$$

$$L = \text{angular momentum}$$

$$m = \text{mass}$$

$$P = \text{power}$$

$$p = \text{momentum}$$

$$r = \text{radius or distance}$$

$$T = \text{period}$$

$$t = \text{time}$$

$$U = \text{potential energy}$$

$$v = \text{velocity or speed}$$

$$W = \text{work done on a system}$$

$$x = \text{position}$$

$$\mu = \text{coefficient of friction}$$

$$\theta = \text{angle}$$

$$\tau = \text{torque}$$

$$\omega = \text{angular speed}$$

$$\alpha = \text{angular acceleration}$$

$$\phi = \text{phase angle}$$

$$\vec{F}_s = -k \Delta \vec{x}$$

$$U_s = \frac{1}{2} k (\Delta x)^2$$

$$x = x_{\max} \cos(\omega t + \phi)$$

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$T_s = 2\pi \sqrt{\frac{m}{k}}$$

$$T_p = 2\pi \sqrt{\frac{\ell}{g}}$$

$$|\vec{F}_G| = \frac{G m_1 m_2}{r^2}$$

$$U_G = -\frac{G m_1 m_2}{r}$$

ELECTRICITY AND MAGNETISM

$$|\vec{F}_E| = \frac{1}{4\pi\epsilon_0} \left| \frac{q_1 q_2}{r^2} \right|$$

$$\vec{E} = \frac{\vec{F}_E}{q}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$E_x = -\frac{dV}{dx}$$

$$\Delta V = -\int \vec{E} \cdot d\vec{r}$$

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

$$U_E = qV = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$\Delta V = \frac{Q}{C}$$

$$C = \frac{\kappa \epsilon_0 A}{d}$$

$$C_p = \sum_i C_i$$

$$\frac{1}{C_s} = \sum_i \frac{1}{C_i}$$

$$I = \frac{dQ}{dt}$$

$$U_C = \frac{1}{2} Q \Delta V = \frac{1}{2} C (\Delta V)^2$$

$$R = \frac{\rho \ell}{A}$$

$$\vec{E} = \rho \vec{J}$$

$$I = Ne v_d A$$

$$I = \frac{\Delta V}{R}$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

$$P = I \Delta V$$

$$A = \text{area}$$

$$B = \text{magnetic field}$$

$$C = \text{capacitance}$$

$$d = \text{distance}$$

$$E = \text{electric field}$$

$$\mathcal{E} = \text{emf}$$

$$F = \text{force}$$

$$I = \text{current}$$

$$J = \text{current density}$$

$$L = \text{inductance}$$

$$\ell = \text{length}$$

$$n = \text{number of loops of wire per unit length}$$

$$N = \text{number of charge carriers per unit volume}$$

$$P = \text{power}$$

$$Q = \text{charge}$$

$$q = \text{point charge}$$

$$R = \text{resistance}$$

$$r = \text{radius or distance}$$

$$t = \text{time}$$

$$U = \text{potential or stored energy}$$

$$V = \text{electric potential}$$

$$v = \text{velocity or speed}$$

$$\rho = \text{resistivity}$$

$$\Phi = \text{flux}$$

$$\kappa = \text{dielectric constant}$$

$$\vec{F}_M = q\vec{v} \times \vec{B}$$

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I$$

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{\ell} \times \hat{r}}{r^2}$$

$$\vec{F} = \int I d\vec{\ell} \times \vec{B}$$

$$B_s = \mu_0 n I$$

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}$$

$$\mathcal{E} = -L \frac{dI}{dt}$$

$$U_L = \frac{1}{2} L I^2$$

ADVANCED PLACEMENT PHYSICS C EQUATIONS

GEOMETRY AND TRIGONOMETRY

Rectangle

$$A = bh$$

Triangle

$$A = \frac{1}{2}bh$$

Circle

$$A = \pi r^2$$

$$C = 2\pi r$$

$$s = r\theta$$

Rectangular Solid

$$V = \ell wh$$

Cylinder

$$V = \pi r^2 \ell$$

$$S = 2\pi r \ell + 2\pi r^2$$

Sphere

$$V = \frac{4}{3}\pi r^3$$

$$S = 4\pi r^2$$

Right Triangle

$$a^2 + b^2 = c^2$$

$$\sin \theta = \frac{a}{c}$$

$$\cos \theta = \frac{b}{c}$$

$$\tan \theta = \frac{a}{b}$$

A = area

C = circumference

V = volume

S = surface area

b = base

h = height

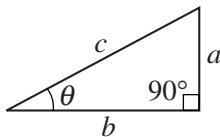
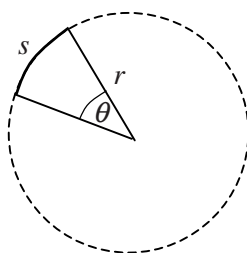
ℓ = length

w = width

r = radius

s = arc length

θ = angle



CALCULUS

$$\frac{df}{dx} = \frac{df}{du} \frac{du}{dx}$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}(\ln ax) = \frac{1}{x}$$

$$\frac{d}{dx}[\sin(ax)] = a \cos(ax)$$

$$\frac{d}{dx}[\cos(ax)] = -a \sin(ax)$$

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}$$

$$\int \frac{dx}{x+a} = \ln|x+a|$$

$$\int \cos(ax) dx = \frac{1}{a} \sin(ax)$$

$$\int \sin(ax) dx = -\frac{1}{a} \cos(ax)$$

VECTOR PRODUCTS

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$|\vec{A} \times \vec{B}| = AB \sin \theta$$

PHYSICS C: ELECTRICITY AND MAGNETISM

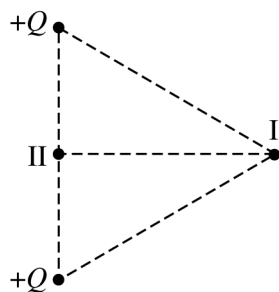
SECTION I

Time—45 minutes

35 Questions

Directions: Each of the questions or incomplete statements below is followed by five suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

Questions 1-2



In the figure above, two small spheres, each with charge $+Q$, are fixed in place at the corners of an equilateral triangle. Point I is at the third corner, and point II is midway between the charges.

1. A small particle with charge $+q$, where $q \ll Q$, is moved from point I to point II at constant speed v by an external force. W_{EXT} is the work done by the external force on the moving charge, and W_{ELEC} is the work done by the electrostatic force. Which of the following correctly identifies the signs of these quantities?

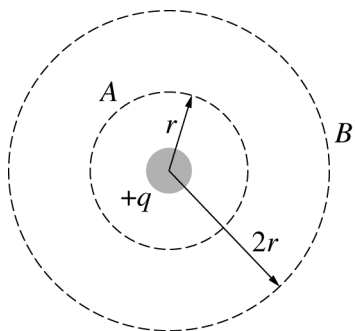
	W_{EXT}	W_{ELEC}
(A)	+	+
(B)	+	–
(C)	–	+
(D)	–	–

- (E) None of the above, since the work done by both the external force and the electrostatic force is zero.

2. Which of the following best describes the relationship between the electric potentials V_I and V_{II} at points I and II, respectively?

- (A) $V_I < V_{II}$
 (B) $V_I = V_{II}$
 (C) $V_I > V_{II}$
 (D) It cannot be determined without knowing the magnitudes of the charges.
 (E) It cannot be determined without knowing the distance between points I and II.

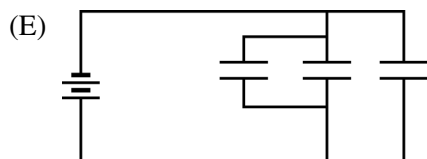
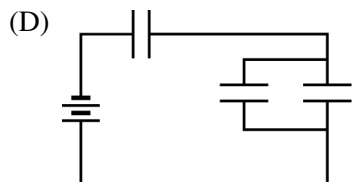
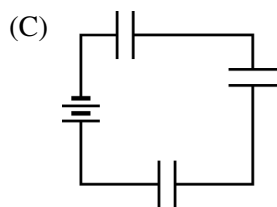
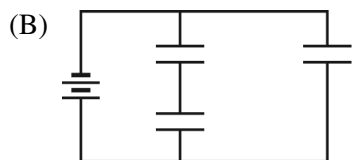
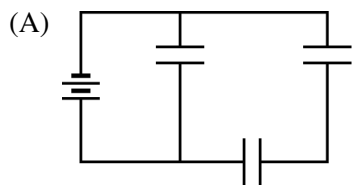
Questions 3-4



A small sphere has a charge $+q$. Spherical Gaussian surfaces A and B are concentric with the sphere, as shown in the figure above. The radii of surfaces A and B are r and $2r$, respectively.

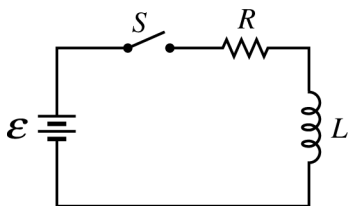
3. The magnitude of the electric flux through A is Φ_A . The magnitude of the electric flux through surface B is Φ_B . The ratio Φ_A/Φ_B is
- (A) $4/1$
(B) $2/1$
(C) $1/1$
(D) $1/2$
(E) $1/4$
4. The magnitude of the electric field at surface A is E_A . The magnitude of the electric field at surface B is E_B . The ratio E_A/E_B is
- (A) $4/1$
(B) $2/1$
(C) $1/1$
(D) $1/2$
(E) $1/4$

5. Each of the following figures shows three identical capacitors connected to a battery. Which arrangement has the greatest equivalent capacitance?



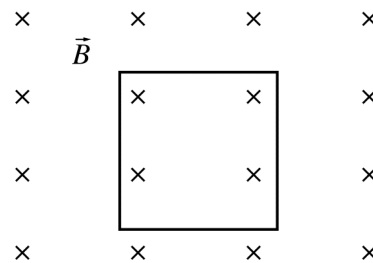
6. A copper wire of length L and diameter D dissipates energy at a rate P_0 when the current in the wire is 10 A. A second copper wire of length L has diameter $2D$. What current in the second wire would dissipate energy at a rate P_0 ?

(A) 2.5 A
 (B) 5.0 A
 (C) 10 A
 (D) 20 A
 (E) 40 A



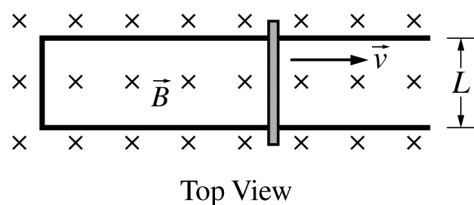
7. Time t is the time it takes the current of an LR circuit with an inductor of inductance L and a resistor of resistance R to reach half of its maximum value. What is the new time if the original inductor is replaced with an inductor of inductance $2L$?

(A) $4t$
 (B) $2t$
 (C) t
 (D) $t/2$
 (E) $t/4$



8. A loop of wire lies in the plane of the page in a region with a uniform magnetic field $\vec{B}$ directed into the page, as shown in the figure above. In which of the following cases, if any, will an emf be induced in the loop at the moment shown in the figure?

(A) The loop is moving toward the right.
 (B) The loop is moving toward the top of the page.
 (C) The loop is moving out of the plane of the paper so that the loop's plane remains perpendicular to the magnetic field.
 (D) The loop is moving into the plane of the paper so that the loop's plane remains perpendicular to the magnetic field.
 (E) An emf cannot be induced in the loop without changing its orientation relative to the magnetic field.



9. A copper rod of resistance R is in electrical contact with a frictionless U-shaped rail of width L and negligible resistance. The rod is pulled to the right at a constant velocity $\vec{v}$. A magnetic field $\vec{B}$ is directed into the page, as shown in the figure above. Under these conditions, the electric power dissipated in the rod is P . If the velocity of the rod is doubled and the magnetic field strength is reduced by half, the power dissipated in the rod is

- (A) $P/4$
 (B) $P/2$
 (C) P
 (D) $2P$
 (E) $4P$

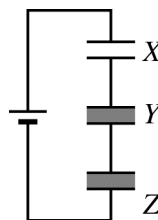


Figure 1

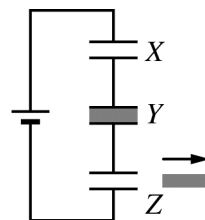
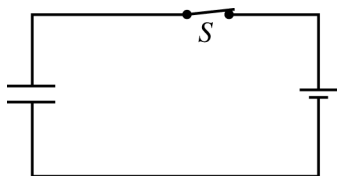


Figure 2

10. Three capacitors are connected in series to an ideal voltage source and charged, as shown in Figure 1 above. The capacitors are identical except that capacitor X has air between its plates, whereas capacitors Y and Z each have a dielectric slab of dielectric constant $\kappa > 1$ between their plates. If the dielectric slab is removed from capacitor Z , as shown in Figure 2, which of the following describes what will happen to the voltage across each capacitor?

Voltage across Capacitor X	Voltage across Capacitor Y	Voltage across Capacitor Z
(A) Increases	Increases	Decreases
(B) Increases	Decreases	Decreases
(C) Increases	Decreases	Increases
(D) Decreases	Increases	Decreases
(E) Decreases	Decreases	Increases

Questions 11-12



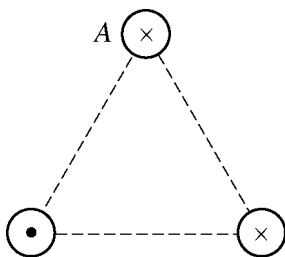
A parallel-plate capacitor connected to a battery is fully charged with the switch S closed, as shown in the circuit above. A slab of dielectric constant $\kappa > 1$ is slowly inserted between the plates of the capacitor.

11. If the switch remains closed when the slab is inserted, what changes, if any, occur?

	Potential Difference <u>Across the Plates</u>	Charge on <u>Positive Plate</u>
(A)	Remains the same	Increases
(B)	Remains the same	Decreases
(C)	Increases	Remains the same
(D)	Decreases	Increases
(E)	Decreases	Remains the same

12. If, instead, the switch is open with the capacitor still fully charged when the slab is inserted, what changes, if any, occur?

	Potential Difference <u>Across the Plates</u>	Charge on <u>Positive Plate</u>
(A)	Remains the same	Increases
(B)	Remains the same	Decreases
(C)	Increases	Remains the same
(D)	Decreases	Increases
(E)	Decreases	Remains the same



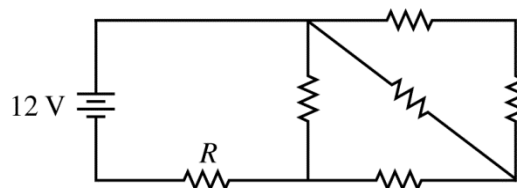
13. Three long wires perpendicular to the page are equidistant from each other, as shown in the cross-sectional view above. Two wires carry current into the page, and the third carries current out of the page. All the currents are equal in magnitude. What is the direction of the net magnetic force on wire A due to the other two wires?

(A) Into the page
 (B) Toward the bottom of the page
 (C) Toward the top of the page
 (D) Toward the left
 (E) Toward the right

14. A beam of particles travels at a right angle to a uniform magnetic field. Particles can be separated into different trajectories based on which of the following properties?

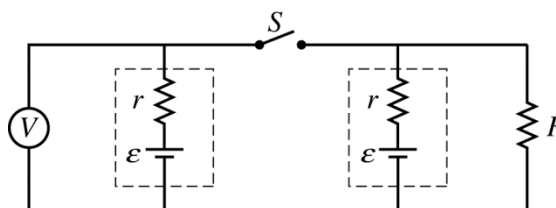
I. The charge of the particle
 II. The mass of the particle
 III. The velocity of the particle

(A) I only
 (B) III only
 (C) I and II only
 (D) I and III only
 (E) I, II and III



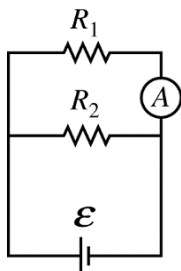
15. Six $2\ \Omega$ resistors are connected to a 12 V battery, as shown in the figure above. What is the current in the resistor labeled R ?

(A) 2.6 A
 (B) 3.7 A
 (C) 4.0 A
 (D) 4.3 A
 (E) 6.0 A



16. A resistor of resistance R is connected in a circuit to two identical batteries. The circuit also contains switch S and ideal voltmeter V , as shown in the figure above. The batteries both have an emf $\mathcal{E}$ and internal resistance r . The reading of the voltmeter is noted with the switch in the open position. Which of the following best represents how the voltmeter reading after the switch is closed compares to the reading before the switch is closed?

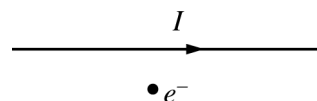
(A) The reading of the voltmeter is the same.
 (B) The reading of the voltmeter is higher.
 (C) The reading of the voltmeter is lower.
 (D) Cannot be determined without knowing the internal resistance of the batteries.
 (E) Cannot be determined without knowing the emf of the batteries.



17. In the circuit shown, $R_1 = 6\ \Omega$, $R_2 = 3\ \Omega$, and the emf $\mathcal{E}$ of the battery is 30 V. The reading of the ammeter A is most nearly

(A) 1.6 A
 (B) 3.3 A
 (C) 5.0 A
 (D) 10 A
 (E) 15 A

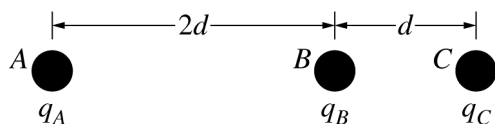
Questions 18-19



An electron is placed near a wire carrying current I , as shown in the figure above, and released from rest. Both the electron and the wire are in the plane of the page.

18. Which of the following is true about the direction of the magnetic field produced by the current at the position of the electron?
- (A) It is toward the top of the page.
 (B) It is toward the bottom of the page.
 (C) It is out of the page.
 (D) It is into the page.
 (E) It has no direction since there is no magnetic field at that point.
19. Which of the following is true about the direction of the initial magnetic force acting on the electron due to the current in the wire?
- (A) It is toward the top of the page.
 (B) It is toward the bottom of the page.
 (C) It is out of the page.
 (D) It is into the page.
 (E) It has no direction because the magnitude of the initial magnetic force on the electron is zero.

Questions 20-22



Three small spheres, A , B , and C , have charges with magnitudes q_A , q_B , and q_C , respectively. The three spheres are aligned along a straight line, as shown in the figure above. At the instant shown, the net force on sphere A is zero.

20. The ratio q_C/q_B is

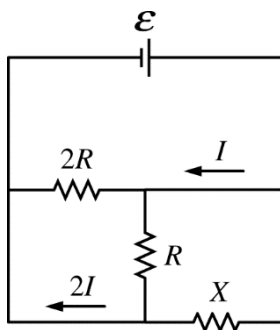
- (A) $9/4$
- (B) $1/1$
- (C) $4/9$
- (D) $1/4$
- (E) $1/9$

21. Which of the following statements must be true of the signs of the charges?

- (A) Only charges q_A and q_B have the same sign.
- (B) Only charges q_A and q_C have the same sign.
- (C) Only charges q_B and q_C have the same sign.
- (D) Charges q_B and q_C have different signs.
- (E) Charges q_A , q_B , and q_C all have the same sign.

22. Which of the following is true about the sign of charge q_A ?

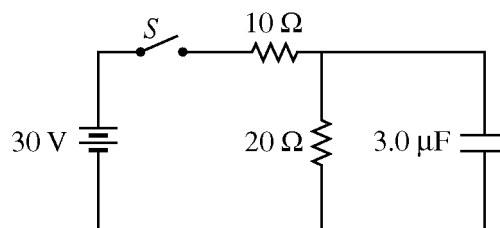
- (A) The sign of charge q_A must be the same as that of q_B .
- (B) The sign of charge q_A must be the same as that of q_C .
- (C) The sign of charge q_A must be the same as that of either q_B or q_C , whichever has the greater magnitude.
- (D) The sign of charge q_A must be the same as that of either q_B or q_C , whichever has the lesser magnitude.
- (E) It is possible that q_A could be either positive or negative.



23. Three resistors are connected to an ideal battery, as shown in the figure above. The battery has an emf $\mathcal{E}$. Two of the resistors have known resistances R and $2R$. The third resistor has unknown resistance X . The current in two of the branches is shown. What is the value of the unknown resistance X ?

(A) $R/5$
 (B) $R/4$
 (C) $R/2$
 (D) R
 (E) $2R$

Questions 24-25



An uncharged $3.0 \mu\text{F}$ capacitor is placed in a circuit with an ideal battery, two resistors, and an open switch S , as shown in the figure above. The switch is then closed.

24. What is the current in the 10Ω resistor immediately after the switch is closed?

(A) Zero
 (B) 1.0 A
 (C) 1.5 A
 (D) 3.0 A
 (E) 10 A

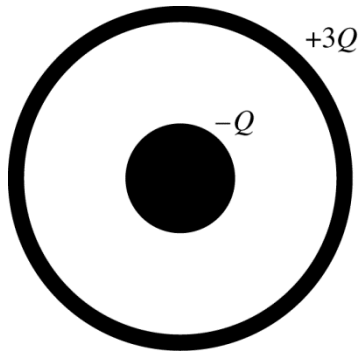
25. What is the current in the 20Ω resistor a long time after the switch is closed?

(A) Zero
 (B) 1.0 A
 (C) 1.5 A
 (D) 3.0 A
 (E) 10 A

26. A parallel-plate capacitor connected to an ideal battery has charge $+Q$ on its top plate. The energy stored in the capacitor is U_C . While the capacitor remains connected to the battery, the separation between the two plates is doubled. Which of the following gives the new charge on the top plate and the new energy stored in the capacitor?

	<u>Charge</u>	<u>Potential Energy</u>
(A)	$+2Q$	$2U_C$
(B)	$+2Q$	$U_C/2$
(C)	$+Q$	U_C
(D)	$+Q/2$	$2U_C$
(E)	$+Q/2$	$U_C/2$

Questions 27-28



A hollow conducting sphere is surrounded by a larger concentric spherical conducting shell, as shown above. The inner sphere has a net charge of $-Q$, and the outer sphere has a net charge of $+3Q$.

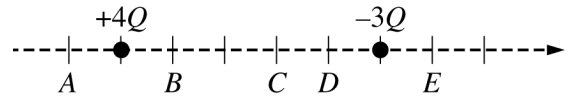
27. What is the net charge on the inner surface of the spherical shell?

(A) $-Q$
 (B) 0
 (C) $+Q$
 (D) $+2Q$
 (E) $+3Q$

28. What is the net charge on the outer surface of the spherical shell?

(A) 0
 (B) $+Q$
 (C) $+2Q$
 (D) $+3Q$
 (E) $+4Q$

Questions 29-30



Two small spheres are arranged along a line and carry charges of $+4Q$ and $-3Q$, as shown in the figure above. The vertical lines are equally spaced.

29. At which of the labeled points does the electric field point toward the right with the smallest magnitude?

(A) A
 (B) B
 (C) C
 (D) D
 (E) E

30. At which of the labeled points does the electric potential have the largest positive value?

(A) A
 (B) B
 (C) C
 (D) D
 (E) E

31. Object A and object B are separated by distance d . Object A has charge $+q$, and object B has charge $-2q$. Object A has a force of magnitude F exerted on it by object B. What are the magnitude and direction of the force exerted on object B?

	<u>Magnitude</u>	<u>Direction</u>
(A)	$F/2$	Away from object A
(B)	F	Away from object A
(C)	$2F$	Away from object A
(D)	F	Toward object A
(E)	$2F$	Toward object A

32. A variable voltage source is connected to an inductor of inductance L . The voltage V as a function of time t is given by the equation $V(t) = \beta t^2$, where β is a constant in units of V/s^2 . The current in the inductor at time $t = 0$ is zero. Which of the following equations gives the magnitude of the current in the inductor as a function of time?

- (A) $I(t) = 0$
 (B) $I(t) = \frac{2\beta}{L}t$
 (C) $I(t) = \frac{\beta}{L}t^2$
 (D) $I(t) = \frac{\beta}{3L}t^3$
 (E) $I(t) = \frac{\beta}{L}t^2 \sin(\omega t)$

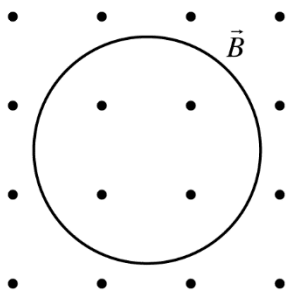


Figure 1

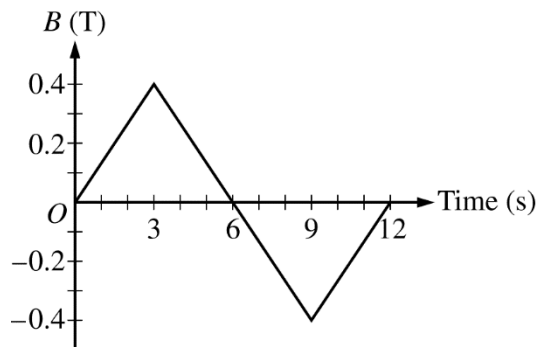
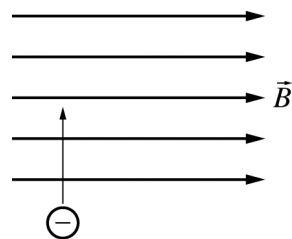


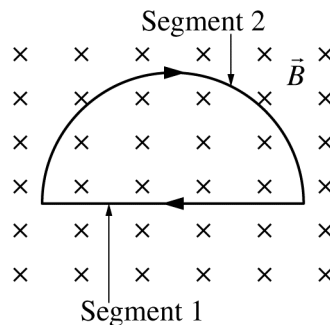
Figure 2

33. A metal wire of resistance $10\ \Omega$ is bent into a circular hoop of radius 0.10 meter and placed in a uniform magnetic field, as shown in Figure 1 above. The magnetic field strength B as a function of time is shown in Figure 2, where positive refers to a magnetic field directed out of the page. What are the magnitude and direction of the current induced in the ring at time $t = 6\text{ s}$?

<u>Magnitude</u>	<u>Direction</u>
(A) 3.8 mA	Clockwise
(B) 3.8 mA	Counterclockwise
(C) 0.42 mA	Clockwise
(D) 0.42 mA	Counterclockwise
(E) 0	No direction



34. A negatively charged ion is moving toward the top of the page when it enters a region of space with a uniform magnetic field $\vec{B}$ directed to the right, as shown above. The direction of the force that the magnetic field exerts on the ion is
- (A) toward the top of the page
 (B) to the right
 (C) to the left
 (D) out of the page
 (E) into the page



35. A semicircular loop with a clockwise current is placed in a uniform magnetic field that is directed into the page, as shown in the figure above. $\vec{F}_1$ is the net force on segment 1, the straight portion of the loop. $\vec{F}_2$ is the net force on segment 2, the curved portion of the loop. Which of the following correctly indicates the directions and relative magnitudes of the forces $\vec{F}_1$ and $\vec{F}_2$?

<u>Direction of $\vec{F}_1$</u>	<u>Direction of $\vec{F}_2$</u>	<u>Magnitudes</u>
(A) Toward the bottom of the page	Toward the top of the page	$ \vec{F}_1 = \vec{F}_2 $
(B) Toward the bottom of the page	Toward the top of the page	$ \vec{F}_1 < \vec{F}_2 $
(C) Toward the bottom of the page	Toward the top of the page	$ \vec{F}_1 > \vec{F}_2 $
(D) Toward the top of the page	Toward the bottom of the page	$ \vec{F}_1 = \vec{F}_2 $
(E) Toward the top of the page	Toward the bottom of the page	$ \vec{F}_1 < \vec{F}_2 $

STOP

END OF ELECTRICITY AND MAGNETISM SECTION I

IF YOU FINISH BEFORE TIME IS CALLED,
YOU MAY CHECK YOUR WORK ON ELECTRICITY AND MAGNETISM SECTION I ONLY.

DO NOT TURN TO ANY OTHER TEST MATERIALS.

MAKE SURE YOU HAVE DONE THE FOLLOWING.

- PLACED YOUR AP NUMBER LABEL ON YOUR ANSWER SHEET
- WRITTEN AND GRIDDED YOUR AP NUMBER CORRECTLY ON YOUR ANSWER SHEET
- TAKEN THE AP EXAM LABEL FROM THE FRONT OF THIS BOOKLET AND PLACED IT ON YOUR ANSWER SHEET

PHYSICS C: ELECTRICITY AND MAGNETISM

SECTION II

Time—45 minutes

3 Questions

Directions: Answer all three questions. The suggested time is about 15 minutes for answering each of the questions, which are worth 15 points each. The parts within a question may not have equal weight. Show all your work in this booklet in the spaces provided after each part.

1. The electric potential in a region of space as a function of position x is given by the equation

$V(x) = \alpha x^2 + \beta x - \gamma$, where $\alpha = 2 \text{ V/m}^2$, $\beta = 7 \text{ V/m}$, and $\gamma = 15 \text{ V}$. All nonelectrical forces are negligible.

- (a) An electron starts at rest at $x = 0$ and travels to $x = 20 \text{ m}$.

i. Calculate the magnitude of the work done on the electron by the electric field during this process.

ii. Calculate the speed of the electron at $x = 20 \text{ m}$.

- (b) Derive an equation for the x -component of the electric field as a function of position x .

(c)

- i. On the axes below, sketch a graph of the acceleration of the electron a as a function of position x .



- ii. On the axes below, sketch a graph of the kinetic energy of the electron K as a function of position x .



- (d) At which of the following locations will an electron that is released from rest move in the negative x direction? Check all that apply.

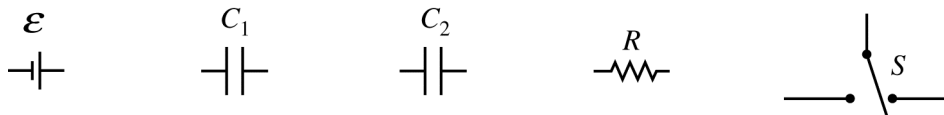
_____ $x = -2$ m _____ $x = +1$ m _____ $x = +3$ m

Justify your answer.

- (e) A charged object, generating its own electric field given by $E(x) = 7$ V/m, is introduced in the region. What is the potential difference from $x = 0$ m to $x = 20$ m caused by the combination of the original electrical potential and the electric field of the charged object?

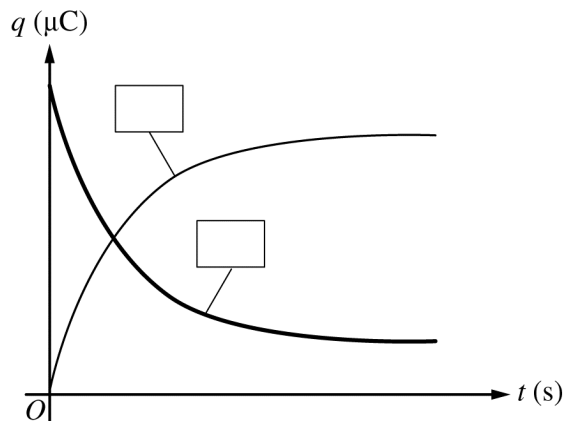
2. A student is to design a circuit using a battery $\mathcal{E}$ with negligible internal resistance, two uncharged capacitors C_1 and C_2 , a resistor R , and a switch S . The circuit should be set up so that when the switch is in one position, the battery will only charge capacitor C_1 , and when in the second position, capacitor C_1 will discharge through capacitor C_2 and resistor R .

- (a) Using the components shown below, draw a circuit diagram that represents a single circuit that will satisfy the criteria outlined above.



The switch S is initially in position to fully charge capacitor C_1 . At time $t = 0$, the switch is moved to the second position. The charge as a function of time for both capacitors after the switch is moved is plotted on the graph below.

- (b) Label each box to indicate which graph corresponds to C_1 and C_2 .



The components of the circuit have the following values: $R = 1000\ \Omega$, $C_1 = 2.0\ \mu\text{F}$, $C_2 = 6.0\ \mu\text{F}$, $\mathcal{E} = 12\ \text{V}$.

(c) Calculate the following for a long time after the switch has been moved to the second position.

i. The charges q_1 and q_2 stored on capacitors C_1 and C_2 , respectively

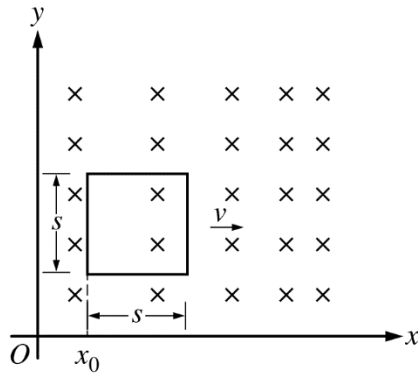
ii. The potential differences V_1 and V_2 across the capacitors C_1 and C_2 , respectively

(d) Calculate the total energy dissipated in resistor R during the time capacitor C_1 discharges.

(e)

i. Write, but do NOT solve, a differential equation that could be used to determine the time constant associated with the discharge of C_1 .

ii. Calculate the time constant associated with the discharge of C_1 .



3. A square loop of conducting wire with resistance R and with sides of length s is located in the xy -plane. The loop is being pushed at constant speed v by an external force of magnitude F_{ext} in the $+x$ -direction through a region with a magnetic field that is directed perpendicular to the xy -plane into the page, as shown in the figure above. The magnitude of the magnetic field as a function of position x is given by the equation $B(x) = cx + B_0$, where x is in meters, c is a positive constant in T/m , and B_0 is a positive constant in teslas. The effects of gravity are negligible.
- (a) Derive an expression for the magnetic flux through the loop when the left side of the loop is at position $x = x_0$. Express your answer in terms of s , c , B_0 , x_0 , and physical constants, as appropriate.
- (b) Derive an expression for the magnitude of the induced current in the loop, if any. Express your answer in terms of R , s , v , c , B_0 , x , and physical constants, as appropriate.

(c) Is the induced current in the loop clockwise, counterclockwise, or zero?

___ Clockwise ___ Counterclockwise ___ Zero

Justify your answer.

(d) When the left side of the loop is at position $x = x_0$ and moving with constant speed v , what is the direction of the net magnetic force, if any, on the loop?

___ Toward the top of the page ___ Out of the page ___ Left

___ Toward the bottom of the page ___ Into the page ___ Right

___ Undefined, because the net magnetic force on the loop is zero.

Justify your answer.

(e) Derive an expression for the magnitude of the external force F_{ext} exerted on the loop to keep it moving at a constant speed when the loop is at position $x = x_0$. Express your answer in terms of R , s , v , c , B_0 , x_0 , and physical constants, as appropriate.

(f) Rank, with 1 being the largest, the magnitude of the magnetic force on the four sides of the loop. If two sides have the same magnetic force, give them the same numerical ranking.

___ Left ___ Right ___ Top ___ Bottom

**Answer Key for AP Physics C: Electricity and Magnetism
Practice Exam, Section I**

Question 1: B	Question 19: E
Question 2: A	Question 20: A
Question 3: C	Question 21: D
Question 4: A	Question 22: E
Question 5: E	Question 23: C
Question 6: D	Question 24: D
Question 7: B	Question 25: B
Question 8: E	Question 26: E
Question 9: C	Question 27: C
Question 10: E	Question 28: C
Question 11: A	Question 29: C
Question 12: E	Question 30: A
Question 13: E	Question 31: D
Question 14: E	Question 32: D
Question 15: B	Question 33: D
Question 16: C	Question 34: D
Question 17: C	Question 35: A
Question 18: D	

2017 AP Physics C: Electricity and Magnetism Question Descriptors and Performance Data

Multiple-Choice Questions

Questions	Topic	Key	% Correct
1	Electrostatics	B	60
2	Electrostatics	A	70
3	Electrostatics	C	53
4	Electrostatics	A	58
5	Electric circuits	E	85
6	Electric circuits	D	43
7	Electromagnetism	B	61
8	Electromagnetism	E	65
9	Electromagnetism	C	71
10	Conductors/capacitors/dielectrics	E	40
11	Conductors/capacitors/dielectrics	A	48
12	Conductors/capacitors/dielectrics	E	51
13	Magnetostatics	E	38
14	Magnetostatics	E	39
15	Electric circuits	B	39
16	Electric circuits	C	26
17	Electric circuits	C	62
18	Magnetostatics	D	77
19	Magnetostatics	E	41
20	Electrostatics	A	70
21	Electrostatics	D	78
22	Electrostatics	E	68
23	Electric circuits	C	40
24	Electric circuits	D	63
25	Electric circuits	B	61
26	Conductors/capacitors/dielectrics	E	54
27	Conductors/capacitors/dielectrics	C	54
28	Conductors/capacitors/dielectrics	C	66
29	Electrostatics	C	42
30	Electrostatics	A	49
31	Electrostatics	D	67
32	Electromagnetism	D	57
33	Electromagnetism	D	34
34	Magnetostatics	D	72
35	Magnetostatics	A	27

2017 AP Physics C: Electricity and Magnetism Question Descriptors and Performance Data

Free-Response Questions

Question	Topic	Mean Score
1	Electrostatics	6.68
2	Electric circuits	5.68
3	Electromagnetism	3.95

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Question 1

15 points total

**Distribution
of points**

(a)

i) 2 points

For using the equation relating the potential difference to the magnitude of the work done on the electron

1 point

$$W = |q||\Delta V| = (1.6 \times 10^{-19} \text{ C}) \left[((2)(20 \text{ m})^2 + (7)(20 \text{ m}) - (15)) - (0 + 0 - 15) \right]$$

For a correct answer with units

1 point

$$W = 1.50 \times 10^{-16} \text{ J} = 940 \text{ eV}$$

ii) 2 points

For using the equation relating the magnitude of the work done on the electron to the speed of the electron

1 point

$$W = \Delta K = \frac{1}{2} m (v_2^2 - v_1^2) = \frac{1}{2} m v^2$$

For correct substitutions

1 point

$$v = \sqrt{\frac{2W}{m}} = \sqrt{\frac{(2)(1.50 \times 10^{-16} \text{ J})}{(9.11 \times 10^{-31} \text{ kg})}}$$

$$v = 1.82 \times 10^7 \text{ m/s}$$

(b) 2 points

For taking the derivative of the electric potential to calculate the electric field

1 point

$$E_x = -\frac{dV}{dx} = -\frac{d}{dx}(2x^2 + 7x - 15)$$

For a correct answer

1 point

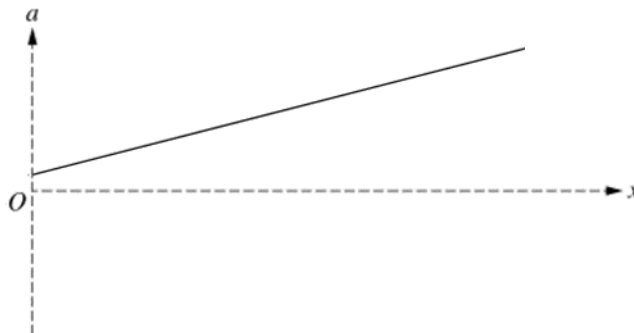
$$E_x = -4x - 7$$

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Question 1 (continued)

**Distribution
of points**

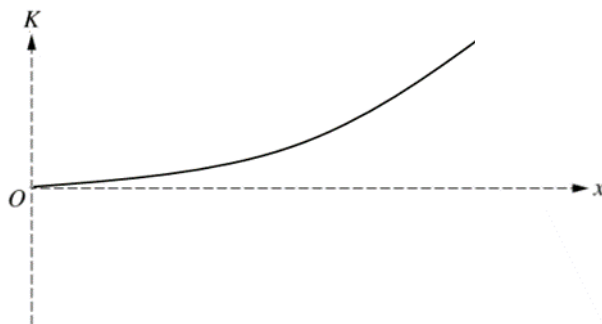
- (c)
i) 2 points



For a straight line for $x > 0$ with non-zero slope
For a line with positive slope and positive y intercept

1 point
1 point

- ii) 2 points



For a graph starting at the origin and extending into the first quadrant
For a concave up curve with $K \geq 0$

1 point
1 point

- (d) 2 points

For selecting only “ $x = -2$ m”
For a correct justification
Example: Electrons accelerate opposite the direction of the electric field. Therefore the electron will accelerate in the negative direction only if the electric field is directed in the positive direction. The electric field is positive only for $x < -1.75$ m.
Note: If incorrect selection is made, justification cannot earn credit.

1 point
1 point

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Question 1 (continued)

**Distribution
of points**

(e) 3 points

Starting with the integration of the net electric field to determine the potential difference

$$V_{20} - V_0 = -\int E \cdot dx$$

For recognizing that $E = E_1 + E_2$

1 point

$$V_{20} - V_0 = -\int E_{net} \cdot dx = -\int (E_1 + E_2) \cdot dx$$

For the substitution of the correct expressions for the electric fields

1 point

$$\Delta V = -\int (E_1 + E_2) dx = -\int [(-4x - 7) + 7] dx = -\int -4x dx = \int_{x=0}^{x=20} 4x dx$$

For a substitution with the correct limits and signs

1 point

$$\Delta V = \left[2x^2 \right]_{x=0}^{x=20} = (2)(20)^2 - (2)(0)^2$$

$$\Delta V = 800 \text{ V}$$

Alternate solution

Alternate points

For correctly recognizing that the potential difference is equal to the sum of potential differences due to each electric field.

1 point

$$\Delta V = \Delta V_1 + \Delta V_2$$

For determining the potential difference of the original field from part (a)

1 point

$$\Delta V_1 = V(20) - V(0) = 925 - (-15) = 940 \text{ V}$$

For correctly determining the potential difference due to the field generated by the charged object

1 point

$$\Delta V_2 = -\int_0^{20} E dx = -E\Delta x = -\left(7 \frac{\text{V}}{\text{m}}\right)(20 \text{ m}) = -140 \text{ V}$$

$$\Delta V = 940 + (-140) = 800 \text{ V}$$

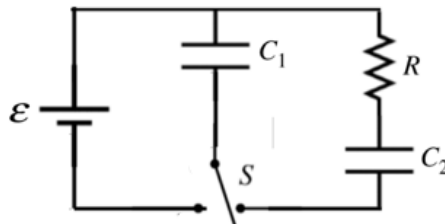
**AP[®] PHYSICS C: ELECTRICITY AND MAGNETISM
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Question 2

15 points total

**Distribution
of points**

(a) 4 points



For drawing a circuit with the battery on a branch by itself

1 point

For having the resistor in series with C_2

1 point

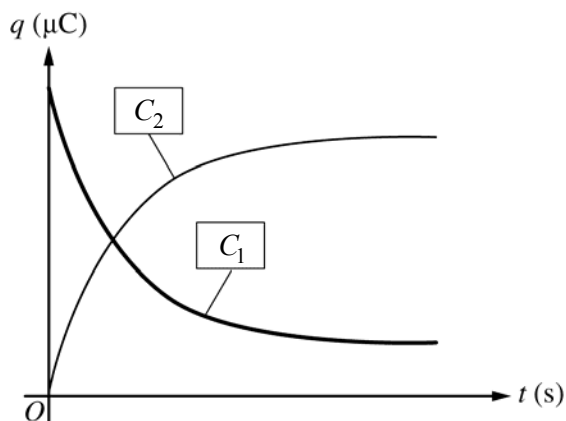
For having the switch positioned so that only C_1 can be charged

1 point

For having the switch positioned so that C_1 can be discharged to C_2

1 point

(b) 1 point



For correctly identifying the two curves

1 point

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Question 2 (continued)

**Distribution
of points**

(c)

i) 3 points

For indicating that the potential difference across the two capacitors is the same

1 point

$$V_1 = V_2$$

$$\frac{q_1}{C_1} = \frac{q_2}{C_2}$$

For correctly relating the final charges on the two capacitors to the original charge on capacitor C_1

1 point

$$Q_{tot} = q_1 + q_2$$

For correctly substituting one of the above equations into the other (note: either equation can be substituted into the other)

1 point

$$\frac{q_1}{C_1} = \frac{q_2}{C_2} \therefore \frac{q_1}{(2.0 \mu\text{F})} = \frac{q_2}{(6.0 \mu\text{F})} \therefore q_2 = 3q_1$$

$$Q_{tot} = q_1 + q_2 \therefore C_1 \mathcal{E} = q_1 + 3q_1 = 4q_1$$

$$(2.0 \mu\text{F})(12 \text{ V}) = 4q_1 \therefore q_1 = 6.0 \mu\text{C}$$

$$q_2 = Q_{tot} - q_1 = 24 \mu\text{C} - 6.0 \mu\text{C} = 18 \mu\text{C}$$

$$q_1 = 6.0 \mu\text{C}$$

$$q_2 = 18 \mu\text{C}$$

ii) 1 points

For correct numerical substitution from part ci into correct equation

1 point

$$V_1 = \frac{q_1}{C_1} = \frac{(6.0 \mu\text{C})}{(2.0 \mu\text{F})} = 3.0 \text{ V}$$

$$V_2 = V_1 = 3.0 \text{ V}$$

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Question 2 (continued)

**Distribution
of points**

(d) 3 points

For indicating that the energy dissipated in the resistor is the difference in the initial and final energy stored in the capacitors

1 point

$$E = \Delta U = U_i - U_f$$

For indicating the final energy stored is the sum of the energy across the two capacitors

1 point

$$E = U_{li} - (U_{1f} + U_{2f}) = \left(\frac{1}{2} C_1 \mathcal{E}^2 \right) - \left(\frac{q_{1f}^2}{2C_1} + \frac{q_{2f}^2}{2C_2} \right)$$

For correctly substituting numerically into the equation above

1 point

$$E = \left(\frac{1}{2} C_1 \mathcal{E}^2 \right) - \left(\frac{q_{1f}^2}{2C_1} + \frac{q_{2f}^2}{2C_2} \right) = \left(\frac{1}{2} \right) (2.0 \text{ } \mu\text{F}) (12 \text{ V})^2 - \left(\frac{(6.0 \text{ } \mu\text{C})^2}{(2)(2.0 \text{ } \mu\text{F})} + \frac{(18 \text{ } \mu\text{C})^2}{(2)(6.0 \text{ } \mu\text{F})} \right)$$

$$E = 108 \text{ } \mu\text{J}$$

(e)

i) 2 point

For an application of Kirchhoff's loop rule to the circuit with the switch in the second position

1 point

$$V_{C1} - V_{C2} - V_R = 0$$

$$\frac{q_1}{C_1} - \frac{q_2}{C_2} - IR = 0$$

For a correct differential equation that could be used to calculate the time constant associated with the discharging of C_1

1 point

$$\frac{Q - q}{C_1} - \frac{q}{C_2} - R \frac{dq}{dt} = 0$$

ii) 1 points

For indicating that the time constant, RC , can be calculated using capacitors in series

1 point

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{6} + \frac{1}{2} = \frac{4}{6}$$

$$C_{eq} = 1.5 \text{ } \mu\text{F}$$

$$\tau = RC = (1000 \text{ } \Omega)(1.5 \text{ } \mu\text{F})$$

$$\tau = 1.5 \text{ ms}$$

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Question 3

15 points total

**Distribution
of points**

(a) 3 points

For substituting the expression for the magnetic field into the integration to calculate the magnetic flux

1 point

$$\Phi = \int \mathbf{B} \cdot d\mathbf{A} = \int (cx + B_0) s dx$$

For integrating the expression above using proper limits or constants of integration

1 point

$$\Phi = s \left[\frac{cx^2}{2} + B_0 x \right]_{x=x_0}^{x=x_0+s} = s \left[\left(\frac{c(x_0+s)^2}{2} + B_0(x_0+s) \right) - \left(\frac{c(x_0)^2}{2} + B_0(x_0) \right) \right]$$

$$\Phi = s \left(\frac{cx_0^2 + 2csx_0 + cs^2}{2} + B_0x_0 + B_0s - \left(\frac{cx_0^2}{2} + B_0x_0 \right) \right)$$

For a correct answer

1 point

$$\Phi = s^2 \left(cx_0 + \frac{1}{2}cs + B_0 \right)$$

(b) 3 points

For using the derivative of the expression from part (a) to calculate the emf

1 point

$$\mathcal{E} = \left| -\frac{d\Phi}{dt} \right| = \frac{d}{dt} \left[s^2 \left(cx + \frac{1}{2}cs + B_0 \right) \right]$$

For applying the chain rule and substituting v for dx/dt

1 point

$$\mathcal{E} = \frac{d\Phi}{dx} \frac{dx}{dt} = \frac{d}{dx} \left[s^2 \left(cx + \frac{1}{2}cs + B_0 \right) \right] v = vs^2 \frac{d}{dx} \left(cx + \frac{1}{2}cs + B_0 \right)$$

$$\mathcal{E} = vs^2 c$$

For dividing by the resistance to calculate the current

1 point

$$I = \frac{\mathcal{E}}{R} = \frac{(vs^2 c)}{R} = \frac{vs^2 c}{R}$$

(c) 2 points

Selecting “Counterclockwise”

For indicating that the magnetic flux inside the loop is increasing with time

1 point

For correctly using Lenz’s law to relate the change in flux to the direction of the induced current

1 point

Example: As the loop moves toward the right, the magnetic field is increasing so the magnetic flux inside the loop is increasing. According to Lenz’s law, the induced current must create a magnetic field that is out of the page to oppose this increasing flux. Therefore, the current in the loop must be counterclockwise.

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Question 3 (continued)

**Distribution
of points**

(d) 3 points

For correctly selecting “Left”

1 point

For indicating that the magnetic forces on the top and bottom parts of the loop are of equal magnitude

1 point

For indicating that the magnetic force on the right side of the loop is greater than the force on the left side of the loop

1 point

Example: The magnitude of the magnetic field is the same for the top and bottom of the loop so the forces on the top and bottom of the loop are equal in magnitude and opposite in direction and thus cancel out. The magnitude of the magnetic field is greater at the right side of the loop than at the left side of the loop thus the net force is in the direction of the force on the right side of the loop and this would be toward the left.

(e) 3 points

For indicating that the magnetic force on the loop is the difference in the force on the left and right sides of the loop

1 point

$$F_{ext} = F_R - F_L = B_R IL - B_L IL$$

For correctly substituting for the magnetic fields into the above equation

1 point

$$F_{ext} = IL(B_R - B_L) = Is[(c(x + s) + B_0) - (cx + B_0)]$$

For correctly substituting current from part (b)

1 point

$$F_{ext} = \left(\frac{vs^2 c}{R} \right) s(cs) = \frac{vs^4 c^2}{R}$$

(f) 1 point

For correctly indicating the correct rankings

1 point

 3 Left 1 Right 2 Top 2 Bottom